Jiaqi Guan

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Summary

Research scientist focused on **foundation models, GenAI, and large-scale ML**. Developed **billion-parameter transformer and diffusion models** on hundreds of GPUs, spanning model architecture, scaling behavior, and efficient training for multimodal generation. Worked on open-source foundation models including **Protenix** and **PXDesign**, with publications in **Cell, NeurIPS, ICML, ICLR, and CVPR**.

Education

University of Illinois Urbana-Champaign — Aug 2019–May 2024
Ph.D. in Computer Science. Advisor: Prof. Jian Peng. — Urbana, IL, USA

Tsinghua University — Aug 2014–Jul 2018
B.E. in Automation. — Beijing, China

Experience

ByteDance, Seed — Jun 2024–Present
Senior Research Scientist — Bellevue, WA

- Worked on model architecture and large-scale training of **Protenix** and **Protenix-v1**, biomolecular foundation models combining transformer architectures and diffusion-based generation, achieving **state-of-the-art performance** for protein complex structure prediction among open-source systems.
- Developed and trained **billion-parameter transformer and diffusion models on 256 GPUs**, designing training strategies (mixed precision, activation checkpointing, custom kernels, DeepSpeed EvoAttention) for efficient and stable large-scale training; achieved **2× speedup**, **30% memory reduction**, and scaled inference from **20k to 60k atoms**.
- Led **scaling and architecture studies** for next-generation models, including depth scaling with recycling embeddings and width scaling with Mixture-of-Experts variants, studying scaling behavior and guiding model design and compute allocation for subsequent large-scale training runs.
- Built **PXDesign**, a generative modeling system integrating diffusion-based generation with large-scale inference and confidence-driven ranking, enabling 10^5-scale candidate generation and screening, achieving state-of-the-art **17–82%** hit rates for protein binder design.
- Drove open-source release of **Protenix** and **PXDesign**, with Protenix reaching **1.7k+** GitHub stars and PXDesign adopted by **10+ academic institutions** with downstream experimental validation.

University of Illinois Urbana-Champaign — Aug 2019–May 2024
Research Assistant, Department of Computer Science — Urbana, IL

- Developed a research line in **geometric deep learning** and **generative modeling**, building equivariant network architectures and diffusion-based methods for structured 3D data, with first-author publications at **ICLR, ICML, and NeurIPS**.
- Designed generative frameworks spanning **diffusion** and **autoregressive modeling**, including formulations for **joint discrete–continuous multimodal generation**.

Tencent, AI Lab May 2020–Aug 2020
NLP Research Intern (Advisor: Liwei Wang) Seattle, WA

- Worked on **large-scale language model pretraining and dialogue generation**, building pretraining and finetuning pipelines over **200M** conversational examples for both **retrieval-based** (poly-encoder) and **generative** (Seq2Seq Transformer) models.
- Explored **reinforcement learning** and **self-play** for dialogue generation, designing reward functions and evaluation metrics to optimize long-horizon behaviors such as sentiment, repetition, and engagement, with improvements on reward-aligned metrics including *DeepMoji* and *question score*.

Carnegie Mellon University Sep 2018–Jan 2019
Research Intern, Robotics Institute (Advisor: Kris Kitani) Pittsburgh, PA

- Developed a **generative model** for structured video forecasting that jointly predicts future **discrete actions** and **continuous trajectories** from visual and scene context.
- Built the model on a normalizing-flow framework for exact likelihood evaluation, and extended it with theoretical analysis for no-regret online learning, resulting in a **CVPR Oral** publication.

Selected Publications

* denotes equal contribution

- Technical Reports and Open-Source Models (**Core Contributor**)
  - Protenix-v1: Toward High-Accuracy Open-Source Biomolecular Structure Prediction
  - Protenix: Advancing Structure Prediction Through a Comprehensive AlphaFold3 Reproduction
  - PXDesign: Fast, Modular, and Accurate De Novo Design of Protein Binders
- Unified Modeling of 3D Molecular Generation via Atomic Interactions with PocketXMol
  Xingang Peng, Ruihan Guo, Fenglin Guo, Ziyi Wang, Jiayu Sun, **Jiaqi Guan**, et al. ***Cell***, 2026
- LinkerNet: Fragment Poses and Linker Co-Design with 3D Equivariant Diffusion
  Jiaqi Guan, Xingang Peng, Peiqi Jiang, Yunan Luo, Jian Peng, Jianzhu Ma. ***NeurIPS (Spotlight)***, 2023
- DecompDiff: Diffusion Models with Decomposed Priors for Structure-Based Drug Design
  Jiaqi Guan*, Xiangxin Zhou*, Yuwei Yang, Yu Bao, Jian Peng, et al. ***ICML***, 2023
- 3D Equivariant Diffusion for Target-Aware Molecule Generation and Affinity Prediction
  Jiaqi Guan*, Wesley Wei Qian*, Xingang Peng, Yufeng Su, Jian Peng, Jianzhu Ma. ***ICLR***, 2023
- Energy-Inspired Molecular Conformation Optimization
  Jiaqi Guan*, Wesley Wei Qian*, Qiang Liu, Wei-Ying Ma, Jianzhu Ma, Jian Peng. ***ICLR***, 2022
- Generative Hybrid Representations for Activity Forecasting with No-Regret Learning
  Jiaqi Guan, Ye Yuan, Kris M. Kitani, Nicholas Rhinehart. ***CVPR (Oral)***, 2020

Technical Skills

Large-Scale ML: Distributed training and inference, mixed-precision, memory optimization, activation checkpointing, custom kernels, inference optimization, scaling studies

Modeling: Foundation models, transformers, diffusion models, structured generation, multimodal modeling, geometric deep learning, reinforcement learning

Programming & Tools: Python, PyTorch, C/C++, Linux, Git, Bash, NumPy/SciPy